

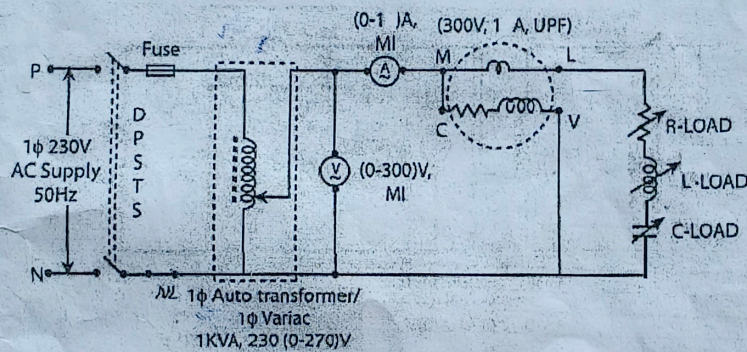
AIM OF THE EXPERIMENTS:-

Calculation of current, voltage and power in series R-L-C circuit excited by single phase AC supply and calculation of power factor.

APPARATUS REQUIRED:-

Sl.no	Instrument/Equipment	Range	Qty.
1	Resistance, inductance & capacitance	Variable type	Per each
2	Ammeter (MI TYPE)	0.5/1A	1
3	Voltmeter (MI TYPE)	0-150,300V	1
4	Wattmeter	0.5/1A,150/300V	1
5	Multimeter	Digital type	1
6	Autotransformer	0-270V,10A	1
7	Connecting wire		As per required

CIRCUIT DIAGRAM:-



THEORY: In the present circuit, resistance R, inductance L and capacitance C are connected in series across a voltage V. Thus the current in all the elements of the circuit is the same, where as voltage across each element would be different. The resultant applied voltage V is the phasor sum of all the voltages across these elements. In such a series circuit, following basic relations hold good.

Voltage across the resistance, $V_R = IR$

Voltage across the Inductance, $V_L = IX_L$

Voltage across the Capacitance, $V_C = IX_C$

Voltage applied across the circuit, $V = IZ$

Power drawn by the circuit, $P = VI \cos \phi$

Power factor of the circuit, $\cos \theta = P/VI$

Where, $X_L = \text{Inductive reactance} = \omega L = 2\pi fL$

$X_C = \text{Capacitance reactance} = 1/\omega C = 1/2\pi fC$

$$Z = \text{Impedance of the circuit} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$I = \text{Current flowing in the circuit} = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\text{The power factor, } \cos\phi = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

The power factor of such a circuit is lagging when $\omega L > 1/\omega C$. In case, $\omega L < 1/\omega C$ the power factor of this circuit will be leading.

PROCEDURE:-

1. Connect the circuit as shown in the diagram.
2. Adjust the rheostat for maximum resistance and the auto transformer to the position of zero-output voltage and switch on the supply.
3. Adjust the voltage across the circuit to about 150 V and note I, V_s , V_L , V_C , V_R and W.
4. Adjust the rheostat for several settings and repeat step 3.
5. Adjust the rheostat to the maximum setting and change the different capacitance value and repeat step 4.
6. Compare the values of phase angle as obtained from the meter readings and from the phasor diagrams. (From the phasor diagrams compute $\cos\theta$ and θ as given in the last two columns of the table). Draw phasor diagrams showing V_R , V_L , V_C , V_s , & I for different sets of readings.

OBSERVATION:-

Sl.No	V_s	I	W	V_R	V_C	V_L	R	L	C	$\cos\theta = \frac{W}{V_s I}$

CONCLUSION:-